

Foundations of Mathematics

Math 558 · Penn State

A comparative study of the three dominant modern approaches to mathematical foundations: **set theory** (ZFC), **category theory**, and **type theory** (Martin-Löf dependent type theory, Homotopy Type Theory, Univalent Foundations). The course examines each framework on its own terms and then asks what it means to “found” mathematics at all.

Lecture slides

- [1 · Some History and Setting the Stage](#) — Bolzano, Dedekind, Cantor, Kronecker, Frege, Russell’s paradox.
- [2 · The Foundational Crisis](#) — paradoxes and consistency, predicativity, the axiom of choice, intuitionism vs. Hilbert’s program.

Notes

- [3 · ZFC as a Foundational Theory](#) — metatheory vs. object theory, the finitistic standpoint, formalizing the metatheory inside ZFC, and the axioms of ZFC.
- [4 · First Consequences of the ZFC Axioms](#) — classes and proper classes, no universal set, the empty set, unordered and ordered pairs, and building the first numbers with Union.
- [5 · Relations and Functions](#) — relations and their terminology, domain/range, functions, Cartesian products via Replacement, isomorphism, quotients by an equivalence relation, and the object-theory/metatheory split.
- [6 · Ordinals](#) — well-founded relations, well-orders and their rigidity, von Neumann ordinals, the well-ordering of ON , Burali-Forti, ω as the least limit ordinal, and every well-order as the order type of a unique ordinal.